

Question 1: what did you do?

Provide a description of what you did. In particular, I want to know what you did to research the subject of the report. In most cases the value of the research is in the journey more than the result, same here, the better/clearer/insightful your write up is the better it scores.

I started this assignment by reading through the provided PowerPoint slides for chapter 6 Risk Management: Assessing Risk and watching the provided lecture video. Once I had a good understanding of the assignment and its expectations, my next step was to generate the table log in excel. I did this by creating a process for assigning the asset ID labels and system ID for all the devices. To create the asset IDs, I identified the different information system elements created, which include people, procedures, data, software, hardware, and networking. For labelling people assets, the ids are all generated with starting "PE - #". The procedural assets are all labelled as "PR - #". The data assets are all labelled as "DT - #". The software and hardware assets are all labelled as "SW - #" and "HW - #" respectively. Lastly, the networking assets are all labelled as "NW - #". The system IDs are all "SN - #" which stands for "Stashi Network."

After I created these processes, I started by identifying the hardware assets. I did this by pulling data from previous classes to analyse work done in previous courses that might be beneficial to review as a part of this assignment. One paper that was done that I had saved from CIDM6340 was a Ransomware Recovery paper, in which I had created lists that included "an inventory of network devices, servers that [would] need to be backed up, devices that [would] need updates, identify passwords that [would] also need to be backed up, and a prioritized list of components for recovery" (Aldakkour, 2024). Another assignment I was able to utilize was from this course – CIDM6341 – the Wireshark assignment. As a part of the research I performed for this assignment, I ran several network mapping scans to identify which assets were assigned which IP and MAC addresses, from which I generated a list of assets that are connected to the network. I used this information from this paper to generate a list of Internet of Things (IoT) devices within my personal network (Stashi, 2025). I then compared the current list to the list from the Ransomware Recovery paper to identify differences. There were several that were noted. Below are the lists from Aldakkour, 2024 and Stashi, 2025:

Ransomware Recovery Paper
(Aldakkour, 2024):

Inventory of Network:

1. iPhone (Jane)
2. iPhone (Austin)
3. MacBook Pro (Jane)
4. Gaming PC (Austin)
5. Dell Laptop (Xcel)
6. Smart TV – Roku (Living Room)
7. Smart TV – Apple (Bedroom)
8. Speaker (Living Room)
9. Xbox
10. Nintendo Switch
11. Playstation
12. Alexa A (Living Room – Offline)
13. Alexa B (Office)
14. Alexa C (bedroom)
15. Eeros Router
16. Eeros Router Extendor
17. Ring Camera (Front Door)
18. Ring Camera (Front)
19. Ring Camera (Back)
20. Ring Camera (Garage)
21. Ring Camera (Street Side)
22. Ring Camera (House Side)
23. Ring Camera Connection Extendor
24. Alarm System

CIDM 6341 - Wireshark and Phishing
Assignment (Stashi, 2025):

Inventory of Network:

1. Eero pro E6 router
2. Blink Ring Camera – camera extender
3. Ring Camera – Front Door
4. Ring Camera - Garage
5. Apple TV (Bedroom)
6. Zhejiang Shenghui Lighting (smart lights)
7. Computer Asus (Austin Gaming Computer)
8. Ring Camera – Front of House
9. Ring Camera – Side of House
10. Ring Camera – Side of House
11. Vizio TV (Living Room TV)
12. Ring Camera - Garden
13. Amazon – Alexa (Office)
14. Mobile Phone - Austin
15. Nintendo – Gaming Console
16. Eero – Wi-Fi Extendor
17. Playstation – Gaming Console
18. Mobile Phone – Jane
19. Macbook Pro (Jane’s Computer)

There were several devices that were listed on the Ransomware inventory list that were not registered through the NMAP scan performed for the Wireshark and Phishing Assignment inventory list. The differing items are the dell laptop (Xcel), Xbox, Alexa A (Living Room), Alexa C (bedroom), Alarm System, and Speaker (Living Room) from the Ransomware list which did not appear on the Wireshark list. I determined the reason for each missing device. The dell laptop (Xcel) is missing from the network scan because I no longer work at Xcel and returned their company equipment on my last day. The Xbox was not on the scan, as it is powered off at the time of the scan; however, it is still a device that is considered a part of the network. Alexa A (Living Room) is currently unplugged and being stored in a shed in the back yard, which is why this Alexa is not appearing on the network scans. Alexa C (bedroom) is also currently unplugged; however, it is being stored in the bedroom. The Alarm system is currently turned off and not in use, hence why it is not connected to the network. Lastly, the living room speaker is connected to the TV physically through a HDMI cable.

Since it is not using Wi-Fi to connect to the TV and rather is using a physical cable to do so which means that it does not have a need to connect to the internet to send signals.

With this comparison complete, I then went through and added devices from the Ransomware list that needed to be accounted for in the current asset inventory in the section(s) it was appropriate to add them. Following this, I then went through each room in the house and identified all hardware devices, to which I found a few more to add to the list. One device is my new work laptop, which connects to the network when I telework. Some others included speakers which use Bluetooth to connect to other devices, my camera which has Wi-Fi connectivity capabilities, and a digital photo frame that connects to Wi-Fi to upload new photos.

The next section of assets that I generate a list for were software assets. To start, I used the list of hardware devices to determine software such as operating systems that would be used by such devices. I then generalized what software would be used by a network or by all devices on a network to communicate, for security measures, and naming devices. Once these were listed, I then went to my iPhone and used the "Screen Time" feature on my systems setting application to determine which apps I use the most and looked at the data from the previous week to do this. I then listed the top 15 apps that were listed.

The next section of assets that I took inventory of were the people assets. I took inventory of this section by referring to various legal documents, such as birth certificates and marriage licenses. I also utilized the family tree records on Ancestry.com to locate documented records of familial connections within my network (this family tree was verified through genetic testing). Ancestry.com includes both mine and my husband's family trees. To determine documented "friends" within the network I referred to my contacts list within my phone, for identifying co-workers, I can access the company directory located on the company's employee site, and to identify acquaintances and distant friends I utilized social media sites such as Facebook and Instagram.

There were several ways in which I determined the procedures section of this assignment. The first was to review my planner, which is where I journal daily. In this journal, there are large sections for each day of the week by week and a section in the front that includes a monthly calendar for each month. In the monthly section, I include bigger events, such as birthdays, parties, trips, etc. In the week section, I write a summary of what I did each day. This often includes a short summary of routines, what I did each day, and any strong feelings that I experienced. Using this was helpful in not only identifying certain routines that I had, but also in seeing the patterns when routines certain routines were being practiced and when others were not. For example, some routines that are noted when they are being performed are when I am in the habit of cooking or working out. When I am not in the routine of doing these items, I document them well even including the time I started the activity, details about it (such as the meal cooked if cooking, or the workout performed). Another way I gathered data about routines was by going through my workday. I have routines that I do before going

to work and after leaving work that are in a different category of compared to the rest of my day outside of work due to the secure nature of the it.

The last section completed was the data section. This section was completed by first identifying all repositories that I have. Once these repositories were identified, they were then reviewed for the type of data being stored within them.

Question 2: what were the results?

Summarize and explain what your findings mean. Provide any printouts of products produced, this could be a report, a map, a table, etc.

ID	Asset Name	Description	Sys ID
NW - 001	Eero pro E6	Router	SN - 001
NW - 002	(Blink) Ring Camera	Ring Camera extender	SN - 002
NW - 003	Ring Camera 1	Front Door (video doorbell)	SN - 003
NW - 004	Ring Camera 2	garage (spotlight)	SN - 004
NW - 005	Apple TV	bedroom TV	SN - 005
NW - 006	Zhejiang Shenghui Lighting	smart lights	SN - 006
NW - 007	Computer – Asus	Austin's gaming computer	SN - 007
NW - 008	Ring Camera 3	front of house camera	SN - 008
NW - 009	Ring Camera 4	side of house camera	SN - 009
NW - 010	Ring Camera 5	side of house camera	SN - 010
NW - 011	Vizio TV	living room TV	SN - 011
NW - 012	Ring Camera 6	garden	SN - 012
NW - 013	Amazon Alexa	Alexa	SN - 013
NW - 014	Mobile Phone 1	iPhone - Austin	SN - 014
NW - 015	Nintendo	gaming console	SN - 015
NW - 016	Eero	Wi-Fi extender	SN - 016
NW - 017	Playstation	playstation	SN - 017
NW - 018	Mobile Phone 2	iPhone - Jane	SN - 018
NW - 019	MacBook Pro	Computer - Jane	SN - 019
NW - 020	ONN Soundbar	Soundbar for living room TV	SN - 020
NW - 021	Amazon Alexa 2	Offline - unplugged and powered off, but stored in the master bedroom	SN - 021
NW - 022	Amazon Alexa 3	Offline - unplugged and powered off, stored in the shed (outside of the house)	SN - 022
NW - 023	Xbox	Offline - plugged in but powered off, stored in living room	SN - 023
NW - 024	Alarm System	security system connected to Ring cameras; powered off/offline	SN - 024
NW - 025	Dell Laptop 1	Jane's work laptop	SN - 025

NW - 026	iPad	jane's iPad Air - offline	SN - 026
NW - 027	Dell Laptop 2	Austin's laptop - offline	SN - 027
NW - 028	Speaker	Bluetooth speaker - Sony	SN - 028
NW - 029	Speaker (JBL)	Bluetooth Speaker - JBL	SN - 029
NW - 030	Apple AirPods	Austins airpod	SN - 030
NW - 031	Heaphones	bluetooth headphones	SN - 031
NW - 032	Canon	Camera - offline with Wi-Fi capabilities	SN - 032
NW - 033	Printer	located in the office - offline while powered off	SN - 033
NW - 034	Digital picture frame	used to store and display photographs	SN - 034
NW - 035	Wi-Fi	uor personal Wi-Fi network	SN - 035
NW - 036	DHCP/DNS Services	service used by a router to assign IP addresses to network devices	SN - 036
NW - 037	Router Firewall Rules	network security device	SN - 037
NW - 038	iCloud	used to backuo data for network devices	SN - 038
HW - 001	Eero pro E6	Router	SN - 001
HW - 002	(Blink) Ring Camera	Ring Camera extender	SN - 002
HW - 003	Ring Camera 1	Front Door (video doorbell)	SN - 003
HW - 004	Ring Camera 2	garage (spotlight)	SN - 004
HW - 005	Apple TV	bedroom TV	SN - 005
HW - 006	Zhejiang Shenghui Lighting	smart lights	SN - 006
HW - 007	Computer – Asus	Austin's gaming computer	SN - 007
HW - 008	Ring Camera 3	front of house camera	SN - 008
HW - 009	Ring Camera 4	side of house camera	SN - 009
HW - 010	Ring Camera 5	side of house camera	SN - 010
HW - 011	Vizio TV	living room TV	SN - 011
HW - 012	Ring Camera 6	garden	SN - 012
HW - 013	Amazon Alexa	Alexa	SN - 013
HW - 014	Mobile Phone 1	iPhone - Austin	SN - 014
HW - 015	Nintendo	gaming console	SN - 015
HW - 016	Eero	Wi-Fi extender	SN - 016
HW - 017	Playstation	playstation	SN - 017
HW - 018	Mobile Phone 2	iPhone - Jane	SN - 018
HW - 019	MacBook Pro	Computer - Jane	SN - 019
HW - 020	ONN Soundbar	Soundbar for living room TV	SN - 020
HW - 021	Amazon Alexa 2	Offline - unplugged and powered off, but stored in the master bedrom	SN - 021
HW - 022	Amazon Alexa 3	Offline - unplugged and powered off, stored in the shed (outside of the house	SN - 022

HW - 023	Xbox	Offline - plugged in but powered off, stored in living room	SN - 023
HW - 024	Alarm System	security system connected to Ring cameras; powered off/offline	SN - 024
HW - 025	Dell Laptop 1	Jane's work laptop	SN - 025
HW - 026	iPad	jane's iPad Air - offline	SN - 026
HW - 027	Dell Laptop 2	Austin's laptop - offline	SN - 027
HW - 028	Speaker	Bluetooth speaker - Sony	SN - 028
HW - 029	Speaker (JBL)	Bluetooth Speaker - JBL	SN - 029
HW - 030	Apple AirPods	Austins airpod	SN - 030
HW - 031	Heaphones	bluetooth headphones	SN - 031
HW - 032	Canon	Camera - offline with Wi-Fi capabilities	SN - 032
HW - 033	Printer	located in the office - offline while powered off	SN - 033
HW - 034	Digital picture frame	used to store and display photographs	SN - 034
HW - 035	Subaru	Jane - car that has bluetooth and Apple carplay capabilities	SN - 039
SW - 001	Wi-Fi	uor personal Wi-Fi network	SN - 035
SW - 002	DHCP/DNS Services	service used by a router to assign IP addresses to network devices	SN - 036
SW - 003	Router Firewall Rules	network security device	SN - 037
SW - 004	Subaru	Jane - car that has bluetooth and Apple carplay capabilities	SN - 039
SW - 005	iCloud	used to backuo data for network devices	SN - 038
SW - 006	MacOS	macOS Sonoma; Operating system for SN - 019	SN - 040
SW - 007	Windows OS	Operating system for SN - 007	SN - 041
SW - 008	iOS	Operating system for SN - 014 and SN - 018	SN - 042
SW - 009	Visio SmartCast OS	Smart TV Operating system	SN - 043
SW - 010	Steam	gaming streaming app for video games	SN - 044
SW - 011	Eero App	software for the eero router	SN - 045
SW - 012	Ring App	software for the Ring security system	SN - 046
SW - 013	Apple TV OS	Operating system for SN - 005	SN - 047
SW - 014	Microsoft Office Suite	includes apps such as microsoft word, excel, presentation, etc	SN - 048
SW - 015	Chrome	application	SN - 049
SW - 016	Instagram	application	SN - 050
SW - 017	Spotify	application	SN - 051
SW - 018	Google Maps	application	SN - 052
SW - 019	Facebook	application	SN - 053
SW - 020	Messages	application	SN - 054
SW - 021	Messenger	application	SN - 055
SW - 022	Zillow	application	SN - 056
SW - 023	DoorDash	application	SN - 057

SW - 024	Clock	application	SN - 058
SW - 025	USAA	application	SN - 059
SW - 026	Education	application	SN - 060
SW - 027	Capital One	application	SN - 061
SW - 028	Duo Mobile	application	SN - 062
SW - 029	Mail	application	SN - 063
SW - 030	Quicken	application (computer)	SN - 064
PE - 001	Kittie Aldakkour	my mom - determined by birth certificate	SN - 065
PE - 002	Constantine Aldakkour	my dad - determined by birth certificate	SN - 066
PE - 003	Christy Stashi	Mother-in-Law - determined by birth certificate	SN - 067
PE - 004	Randall Stashi	Father-in-law - determined by birth certificate	SN - 068
PE - 005	William Aldakkour	my brother - determined by family tree/ancestry	SN - 069
PE - 006	Austin Stashi	Husband - determined by marriage certificate	SN - 070
PE - 007	Extended Family	Ancestry.com/Family Tree	SN - 071
PE - 008	Friends	Contacts	SN - 072
PE - 009	Co-Workers	Work Directory	SN - 073
PE - 010	Aquaintances/Distant Friends	Social Media such as Facebook and Instagram	SN - 074
PE - 011	Jane Stashi	me	SN - 075
PR - 001	Morning Routine	routine used to get up in the morning	SN - 076
PR - 002	Driving Routine	routine performed before driving	SN - 077
PR - 003	Work Routine (Arrival)	routine performed before entering work	SN - 078
PR - 004	Work Routine (Departure)	routine performed before leaving work	SN - 079
PR - 005	Home Arrival Routine	routine performed upon arriving home	SN - 080
PR - 006	Cooking Routine	routine performed before, during, and after cooking	SN - 081
PR - 007	Workout Routine	routine performed before, during, and after working out	SN - 082
PR - 008	Homework Routine	routine performed before, during, and after working on and/or completing homework	SN - 083
PR - 009	Bookkeeping Routine	routine performed before, during, and after performing personal bookkeeping	SN - 084
PR - 010	Evening Routine	routine performed before going to bed	SN - 085
DT - 001	MacBook Pro	Contains prior year tax returns, photos, school documents (assignments, powerpoints, notes, etc.), Quicken (financial backup data), medical documents, bills, work	SN - 019

		documents (digital offer letters, resumes, etc.), passwords, gaming data	
DT - 002	Mobile Phone 2	Contains photos, school documents (assignments, powerpoints, notes, etc.), Quicken, app/usage data, medical documents (through BSA app), bills (through financial apps), passwords, banking information, PII	SN - 018
DT - 003	Mobile Phone 1	Contains photos, school documents (assignments, powerpoints, notes, etc.), Quicken, app/usage data, medical documents (through BSA app), bills (through financial apps), passwords, banking information, PII	SN - 014
DT - 004	Computer – Asus	Austin's computer Contains photos, school documents (assignments, powerpoints, notes, etc.), bills, passwords, gaming data	SN - 007
DT - 005	iCloud	Stores photos, documents, backup data for iPhones and Macbook	SN - 038
DT - 006	Gun Safe	Contains important records, such as birth certificates, social security cards, marriage certificate, etc.	SN - 086
DT - 007	BackPack (Blue)	Contains records such as old bank statements, tax documents, school documents (such as SAT and ACT scores and report cards, Insurance cards, legal documents, medical documents, offer letters for jobs, school documentation, etc.	SN - 087
DT - 008	Grey box 1	Austin's records	SN - 088
DT - 009	Grey box 2	Austin's records	SN - 089
DT - 010	Plastic Basket	Mail received	SN - 090
DT - 011	Moving Box 1	Contains school work and notes from my undergrad degree	SN - 091
DT - 012	Moving Box 2	Contains school work and notes from my undergrad degree	SN - 092
DT - 013	Work laptop bag	my laptop bag used for transporting my work laptop	SN - 093
DT - 014	Work backpack	Austin's work backpack used for storing items taken to and from work	SN - 094
DT - 015	USB Drive	USB containing our wedding photos from Scotland	SN - 095
DT - 016	SIM Memory Card	The memory card for the canon camera (item SN - 032)	SN - 096
DT - 017	Social Media Accounts	name, username, passwords, city lived in, jobs (previous), family members, school attended, a link to my Linked In profile, where	SN - 097

		I'm from, who my friends are/who follows me, reels I've saved, people I follow, places I've been recently, things I've done recently, events I'm interested in	
DT - 018	Subaru	Insurance documents, parking passes (old and expired)	SN - 039
DT - 019	Wallet	Drivers license, credit cards	SN - 098

For results, discuss how your inventory translates into attack surface.

The above-mentioned inventory translates into attack surfaces in many ways. Each type of information system components exposes the network to a different kind of security threat. The initial threat level lies with the people who are a part of the network. The people act as an attack surface as bad players can gather information from through the things said online, tracking behavioural patterns, or tricking them into giving up information through phishing attempts or other social engineering tactics. People also act as cybersecurity threats when they fail to create and implement appropriate procedures or when they fail to adhere to procedures. That leads to the next system component, which is a procedure. Procedures can act as a security threat if they are improperly written, poorly implemented, or not properly followed.

More system components that create a broader attack surface include components that interact with internal and external systems. These include software, hardware, and networking components. Software components are perhaps the most vulnerable attack surfaces within a network. Software components have direct access to the data, which is the target looking to be acquired. They are also vulnerable in more ways than one to attacks. They are vulnerable to internal threats through people who have access to the network that will fall for phishing attempts or people who do not follow security policies properly which then expose the network unnecessarily. Furthermore, software is vulnerable to other attacks such as brute force and packet sniffing attacks.

The hardware components are another component which has a larger attack surface area vulnerable to attacks. Hardware devices can be used to store data and are vulnerable to physical attacks as well as virtual attacks. For example, a hardware device such as a phone or laptop which contains data can be stolen as one form of attack or destroyed as another. Should data on these devices not properly be backed up, it would be lost. Hardware often contains software on them which adds to the vulnerability.

Networking components are the last component to discuss which create an attack surface and generate vulnerabilities within a network. Network components are often a point of entry for a bad actor to use to gain entry into a network. Network devices, or IoT devices, can be numerous in a network, and each individual device creates a new vulnerability as each individual device acts as a separate point of entry that must be secured individually. There are often IoT devices within a network that have limited built-in security features which makes it more vulnerable to attack. By finding a weaker IoT device to hack, such as the speakers or

smart lights, a bad actor can infiltrate the network and gain access to more important devices such as a computer or phone.

Question 3: what did you learn?

I can't tell you what you learned. What I will say is that I reward insight. Insights are aha moments (a term in use long before Oprah wanted to copyright it). If you see new ways of doing things, new insights to your thought processes, potential future applications be they personal or work related, crossovers to other topics, these are what I reward more than just telling me you learned lots. I expect you to learn lots, but it isn't till you explain where and what that I see that you really did. Ok, so sometimes you don't learn much. I'll still grade this area high if you tell me why, what you know, how this works on what you've done in the past, etc. Sometimes when you start doing this you see that what you've learned is reinforcing what you've done and sometimes you even have small aha moments. Bottom line is to be reflective, think a few minutes or overnight about what you've done and how it fits into your nomological net (your personal set of knowledge base structure, those theories and beliefs that guide how you evaluate and use knowledge). Then write the section, when I see this done, I always score the section higher.

I learned through performing this assignment that my attack surface is much larger than I originally anticipated. There are many more components within my network to consider than I would have initially thought to include. This includes all the people outside of my household such as friends and family, which I would not have thought to include as potential security risks even though they are. Furthermore, the amount of IoT devices that are within the network is broader than I thought, especially the IoT devices that would be easier to hack into which could be used to gain access from the network was higher than anticipated. These devices include the onn soundbar, the digital picture frame, and the fact that there is an Alexa still active within the house (I thought they had all been decommissioned). Because of the broader network range that we have, there is also a higher risk to attacks. There are more attack surfaces and vulnerabilities than anticipated because of this. There is also a broader range of threats than I thought there could be. I did not consider my hardware as being as vulnerable and at risk to attacks as it is, however, it is vulnerable to many types of threats including being stolen or accidentally or purposely destroyed. This would lead to a loss of data that could be detrimental for me personally. Furthermore, there are other people who have access to the network, such as my husband Austin and my mother-in-law Christy (when she comes to visit from Austin), but also myself. These people create vulnerabilities, as they can expose the network to bad actors by falling for social engineering tactics. Overall, this lesson helped me to realize the extent of my network and data, as well as exposed many vulnerabilities I would not have thought about without a broader knowledge and understanding of the information components that surround me.

References:

- [1] Aldakkour, J. A. (2024). *Ransomware Recovery Assignment*. Unpublished manuscript. College of Business, West Texas A&M University.
- [2] Stashi, J.A. (2025). *CIDM6341 – Wireshark and Phishing Assignment*. Unpublished manuscript. College of Business, West Texas A&M University.